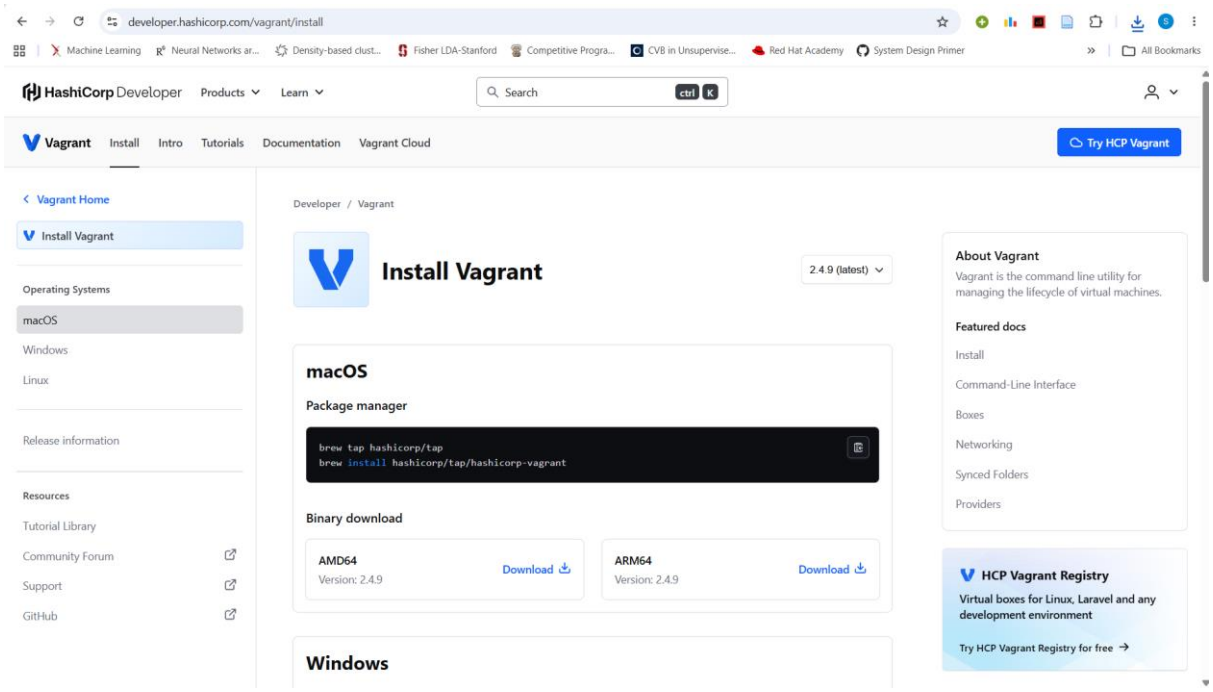
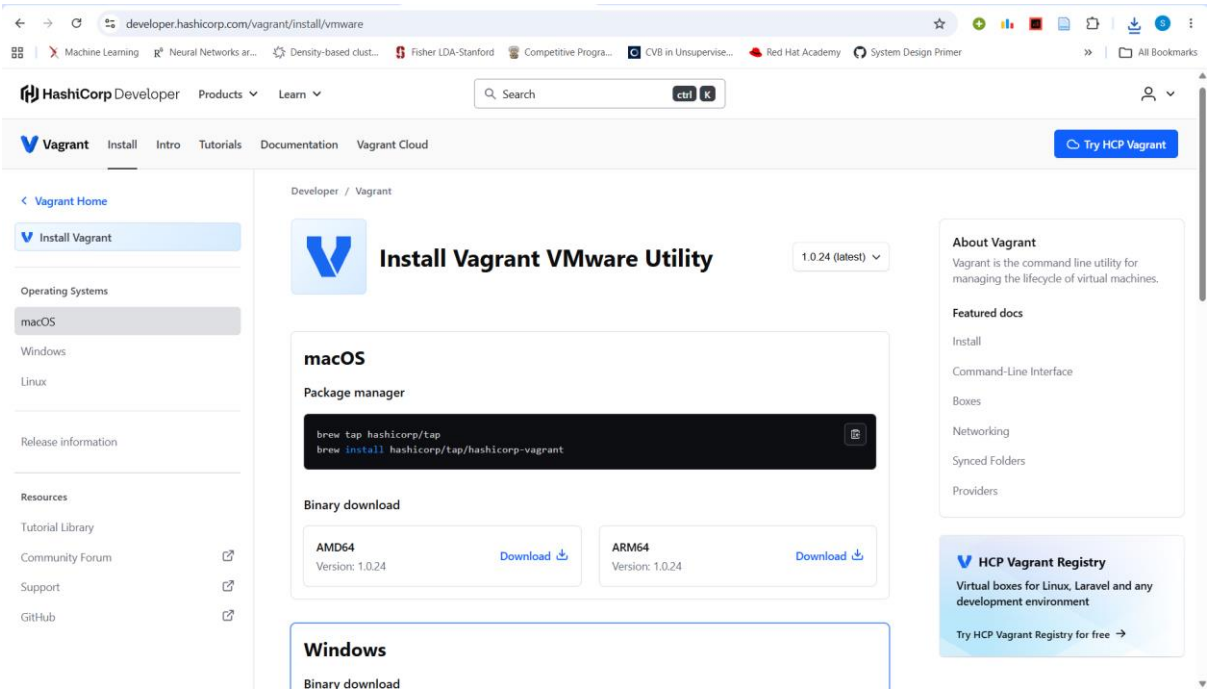
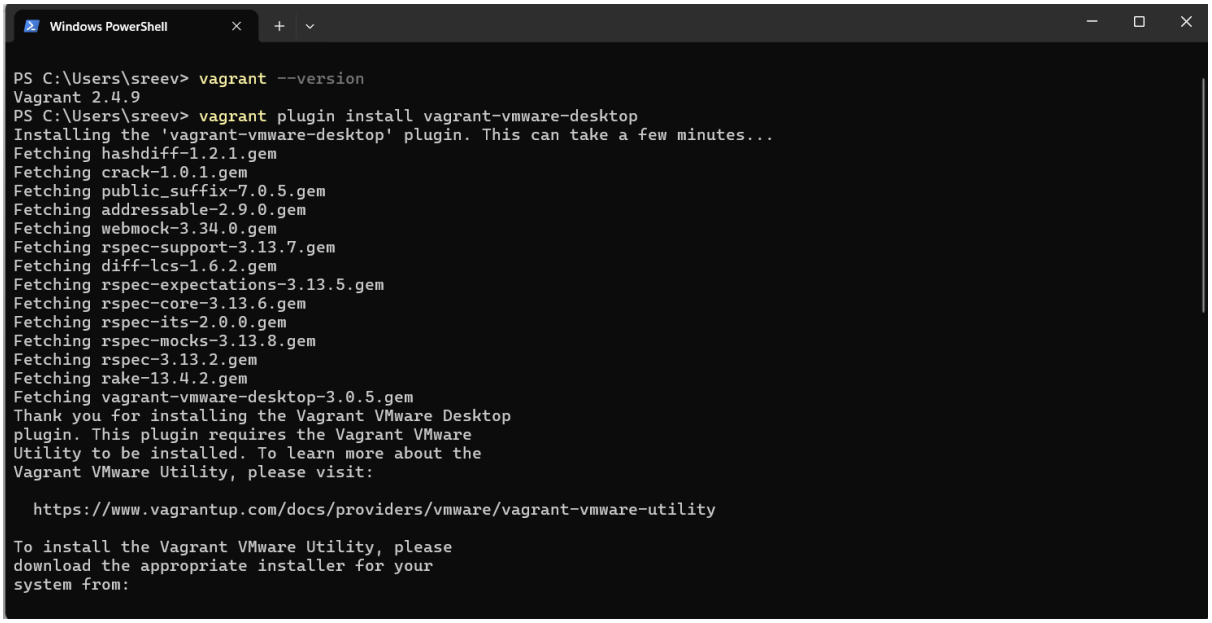
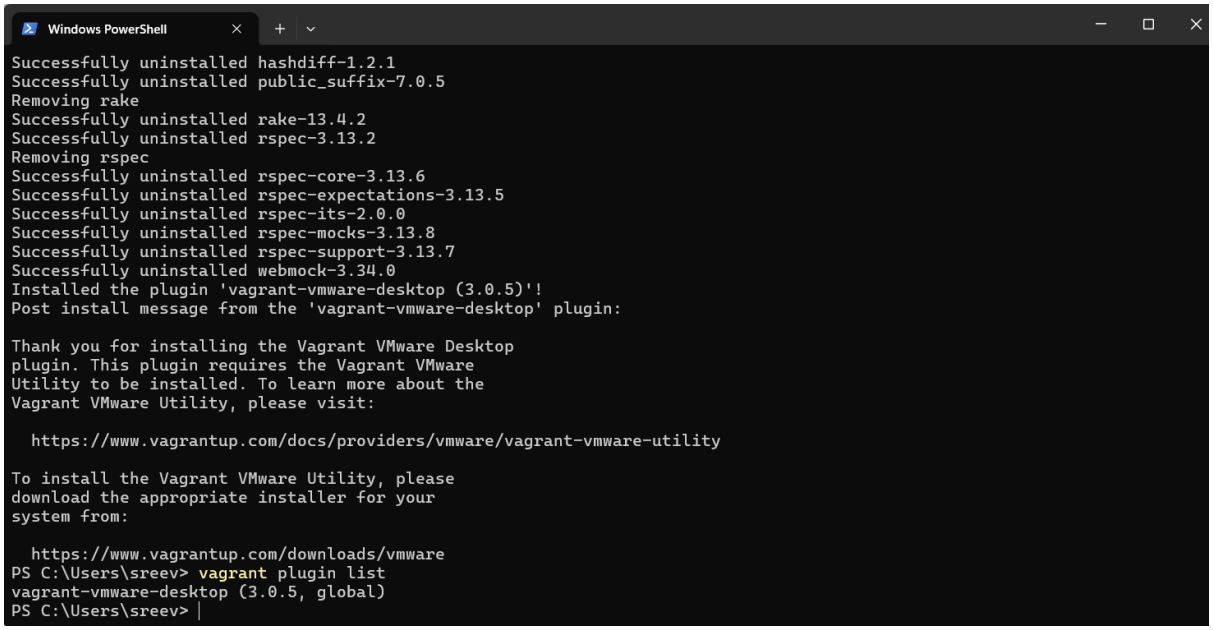
# Practical 4: Vagrant Fundamentals and Ansible Integration

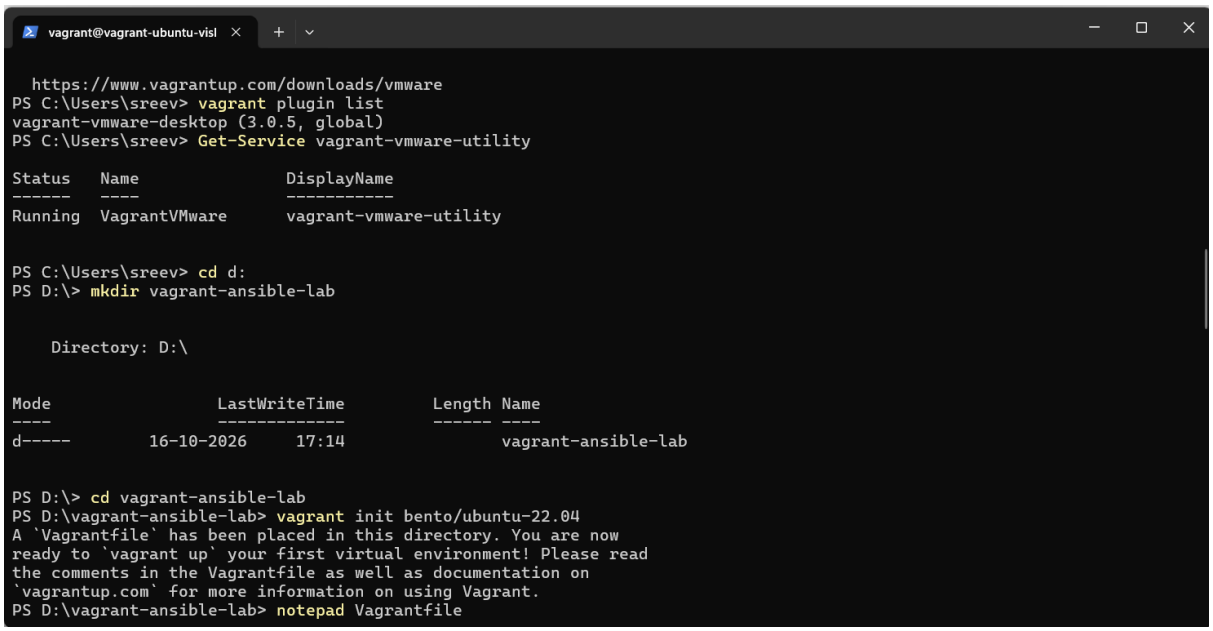
## Vagrant-Based Ubuntu VM Provisioning with VMware Workstation, Ansible Local Configuration, and Nginx Service Management

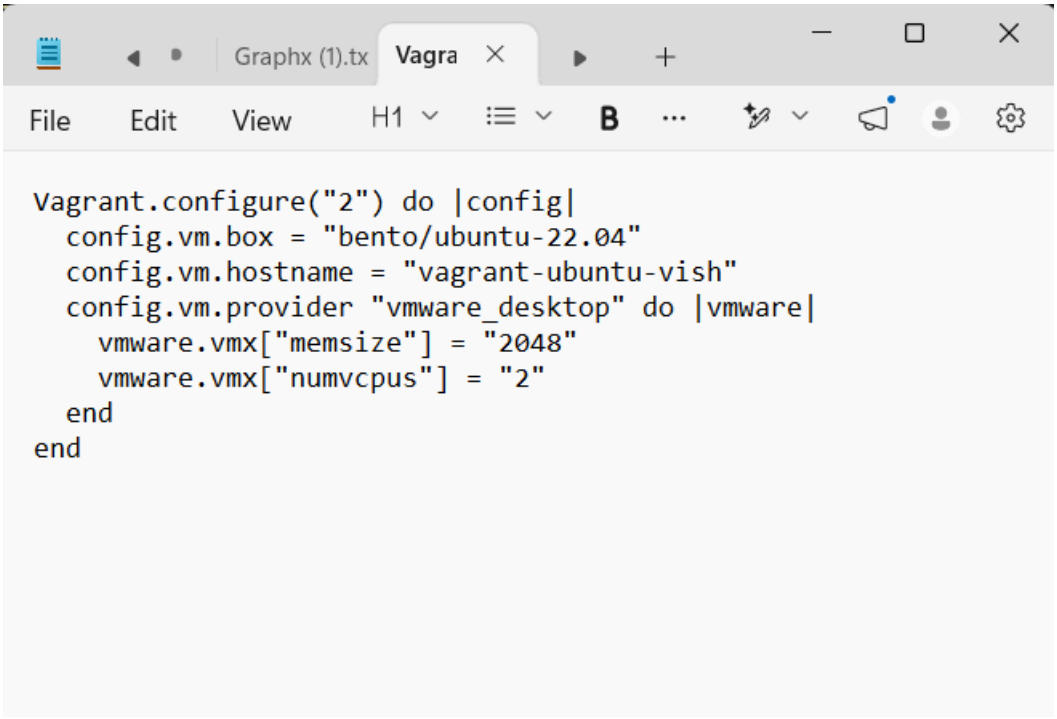


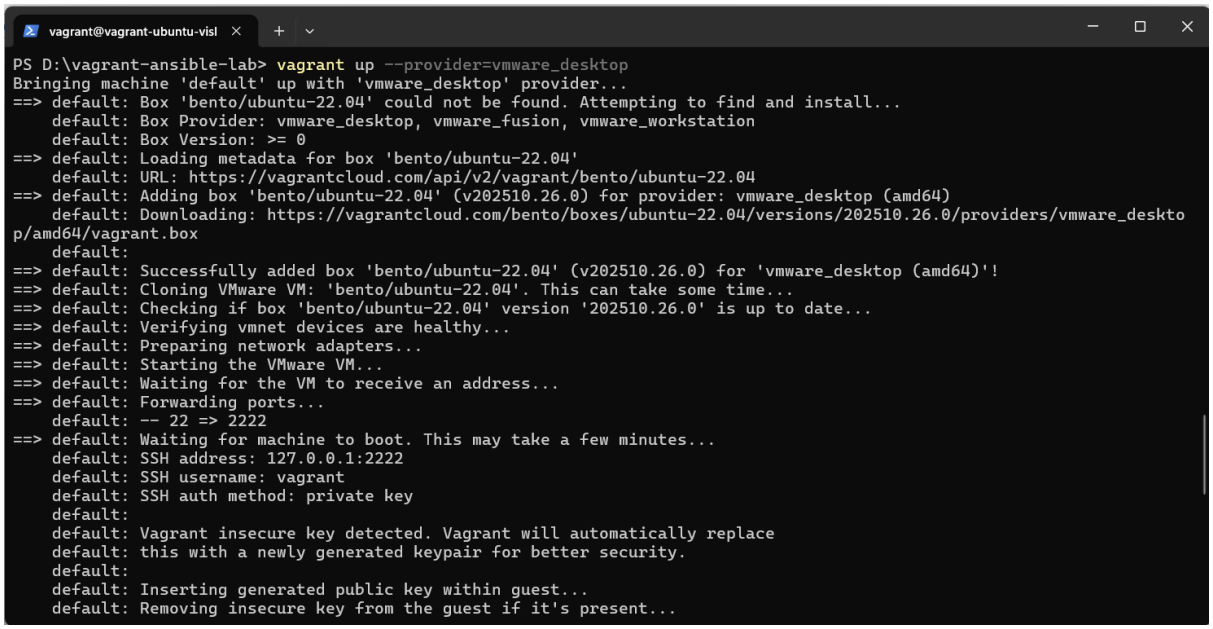

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1.0.24 (latest)
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Package manager
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brew install hashicorp/tap/hashicorp-vagrant
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Version: 1.0.24
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About Vagrant
Vagrant is the command line utility for managing the lifecycle of virtual machines.
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Providers
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Virtual boxes for Linux, Laravel and any development environment
Try HCP Vagrant Registry for free

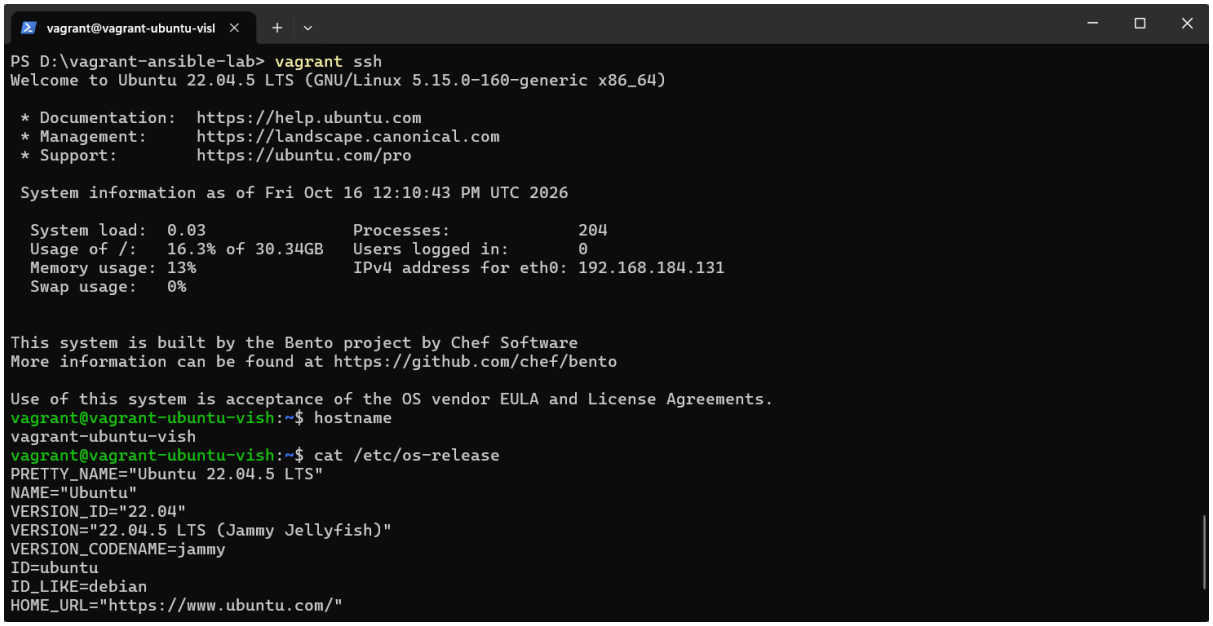

Windows PowerShell
PS C:\Users\sreev> vagrant --version
Vagrant 2.4.9
PS C:\Users\sreev> vagrant plugin install vagrant-vmware-desktop
Installing the 'vagrant-vmware-desktop' plugin. This can take a few minutes...
Fetching hashdiff-1.2.1.gem
Fetching crack-1.0.1.gem
Fetching public_suffix-7.0.5.gem
Fetching addressable-2.9.0.gem
Fetching webmock-3.34.0.gem
Fetching rspec-support-3.13.7.gem
Fetching diff-lcs-1.6.2.gem
Fetching rspec-expectations-3.13.5.gem
Fetching rspec-core-3.13.6.gem
Fetching rspec-its-2.0.0.gem
Fetching rspec-mocks-3.13.8.gem
Fetching rspec-3.13.2.gem
Fetching rake-13.4.2.gem
Fetching vagrant-vmware-desktop-3.0.5.gem
Thank you for installing the Vagrant VMware Desktop
plugin. This plugin requires the Vagrant VMware
Utility to be installed. To learn more about the
Vagrant VMware Utility, please visit:
https://www.vagrantup.com/docs/providers/vmware/vagrant-vmware-utility
To install the Vagrant VMware Utility, please
download the appropriate installer for your
system from:


Windows PowerShell
Successfully uninstalled hashdiff-1.2.1
Successfully uninstalled public_suffix-7.0.5
Removing rake
Successfully uninstalled rake-13.4.2
Successfully uninstalled rspec-3.13.2
Removing rspec
Successfully uninstalled rspec-core-3.13.6
Successfully uninstalled rspec-expectations-3.13.5
Successfully uninstalled rspec-its-2.0.0
Successfully uninstalled rspec-mocks-3.13.8
Successfully uninstalled rspec-support-3.13.7
Successfully uninstalled webmock-3.34.0
Installed the plugin 'vagrant-vmware-desktop (3.0.5)'!
Post install message from the 'vagrant-vmware-desktop' plugin:
Thank you for installing the Vagrant VMware Desktop
plugin. This plugin requires the Vagrant VMware
Utility to be installed. To learn more about the
Vagrant VMware Utility, please visit:
https://www.vagrantup.com/docs/providers/vmware/vagrant-vmware-utility
To install the Vagrant VMware Utility, please
download the appropriate installer for your
system from:
https://www.vagrantup.com/downloads/vmware
PS C:\Users\sreev> vagrant plugin list
vagrant-vmware-desktop (3.0.5, global)
PS C:\Users\sreev>

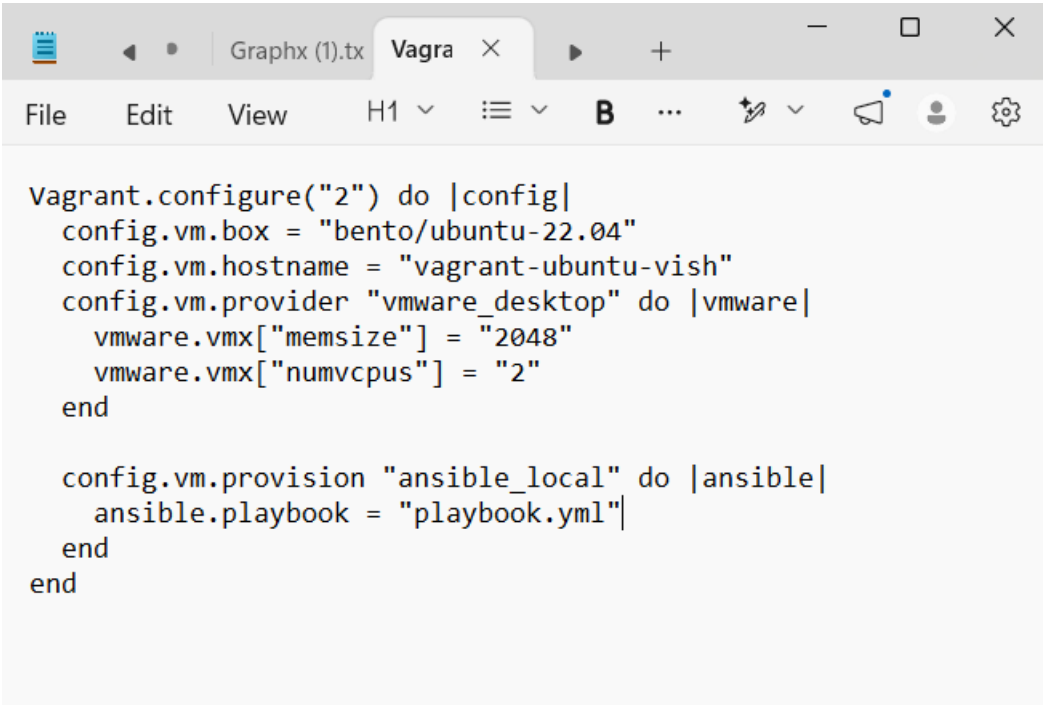

vagrant@vagrant-ubuntu-visl
https://www.vagrantup.com/downloads/vmware
PS C:\Users\sreev> vagrant plugin list
vagrant-vmware-desktop (3.0.5, global)
PS C:\Users\sreev> Get-Service vagrant-vmware-utility
Status Name DisplayName
Running VagrantVMware vagrant-vmware-utility
PS C:\Users\sreev> cd d:
PS D:\> mkdir vagrant-ansible-lab
Directory: D:\
Mode LastWriteTime Length Name
d----- 16-10-2026 17:14 vagrant-ansible-lab
PS D:\> cd vagrant-ansible-lab
PS D:\vagrant-ansible-lab> vagrant init bento/ubuntu-22.04
A `Vagrantfile` has been placed in this directory. You are now
ready to `vagrant up` your first virtual environment! Please read
the comments in the Vagrantfile as well as documentation on
`vagrantup.com` for more information on using Vagrant.
PS D:\vagrant-ansible-lab> notepad Vagrantfile


Graphx (1).tx
Vagra
File
Edit
View
H1
Vagrant.configure("2") do |config|
config.vm.box = "bento/ubuntu-22.04"
config.vm.hostname = "vagrant-ubuntu-vish"
config.vm.provider "vmware_desktop" do |vmware|
vmware.vmx["memsize"] = "2048"
vmware.vmx["numvcpus"] = "2"
end
end

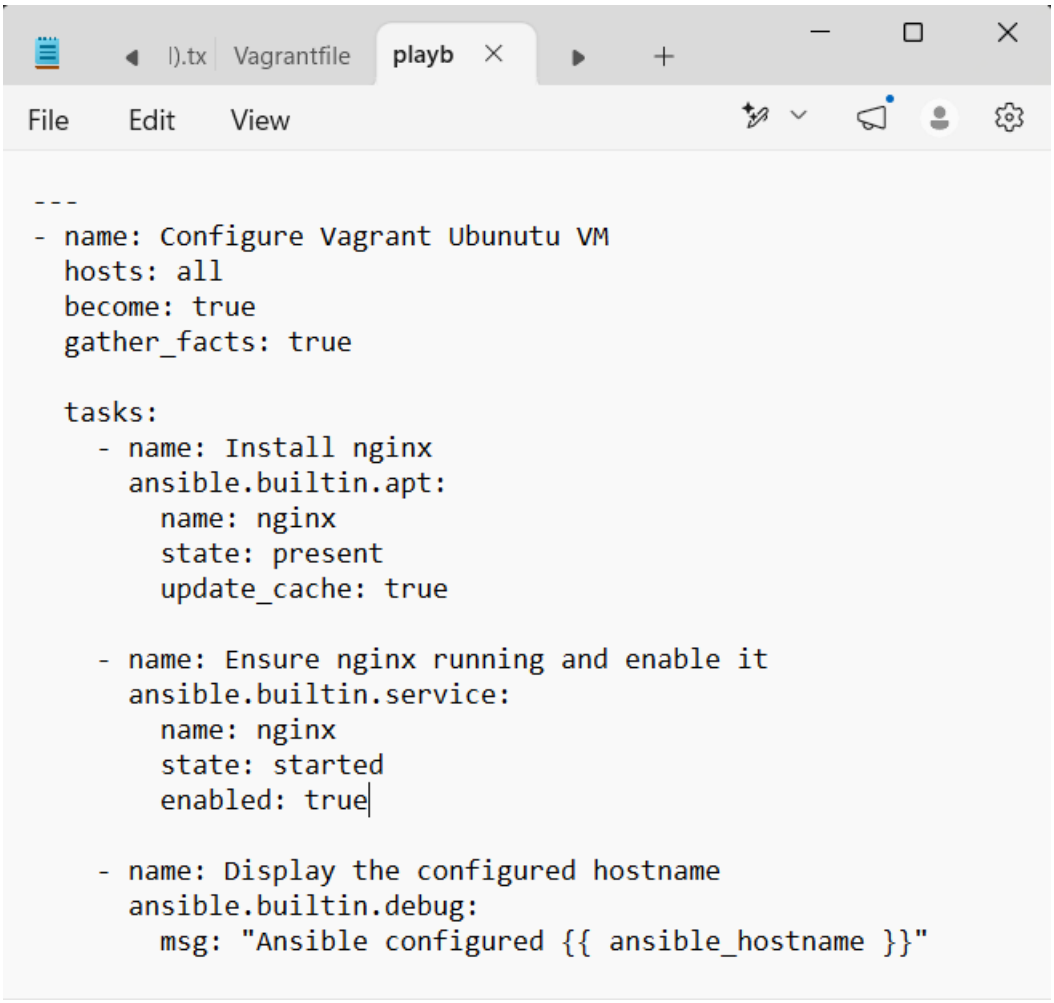

vagrant@vagrant-ubuntu-visl
PS D:\vagrant-ansible-lab> vagrant up --provider=vmware_desktop
Bringing machine 'default' up with 'vmware_desktop' provider...
==> default: Box 'bento/ubuntu-22.04' could not be found. Attempting to find and install...
default: Box Provider: vmware_desktop, vmware_fusion, vmware_workstation
default: Box Version: >= 0
==> default: Loading metadata for box 'bento/ubuntu-22.04'
default: URL: https://vagrantcloud.com/api/v2/vagrant/bento/ubuntu-22.04
==> default: Adding box 'bento/ubuntu-22.04' (v202510.26.0) for provider: vmware_desktop (amd64)
default: Downloading: https://vagrantcloud.com/bento/boxes/ubuntu-22.04/versions/202510.26.0/providers/vmware_desktop/amd64/vagrant.box
default:
==> default: Successfully added box 'bento/ubuntu-22.04' (v202510.26.0) for 'vmware_desktop (amd64)'!
==> default: Cloning VMware VM: 'bento/ubuntu-22.04'. This can take some time...
==> default: Checking if box 'bento/ubuntu-22.04' version '202510.26.0' is up to date...
==> default: Verifying vmnet devices are healthy...
==> default: Preparing network adapters...
==> default: Starting the VMware VM...
==> default: Waiting for the VM to receive an address...
==> default: Forwarding ports...
default: -- 22 => 2222
==> default: Waiting for machine to boot. This may take a few minutes...
default: SSH address: 127.0.0.1:2222
default: SSH username: vagrant
default: SSH auth method: private key
default:
default: Vagrant insecure key detected. Vagrant will automatically replace
default: this with a newly generated keypair for better security.
default:
default: Inserting generated public key within guest...
default: Removing insecure key from the guest if it's present...



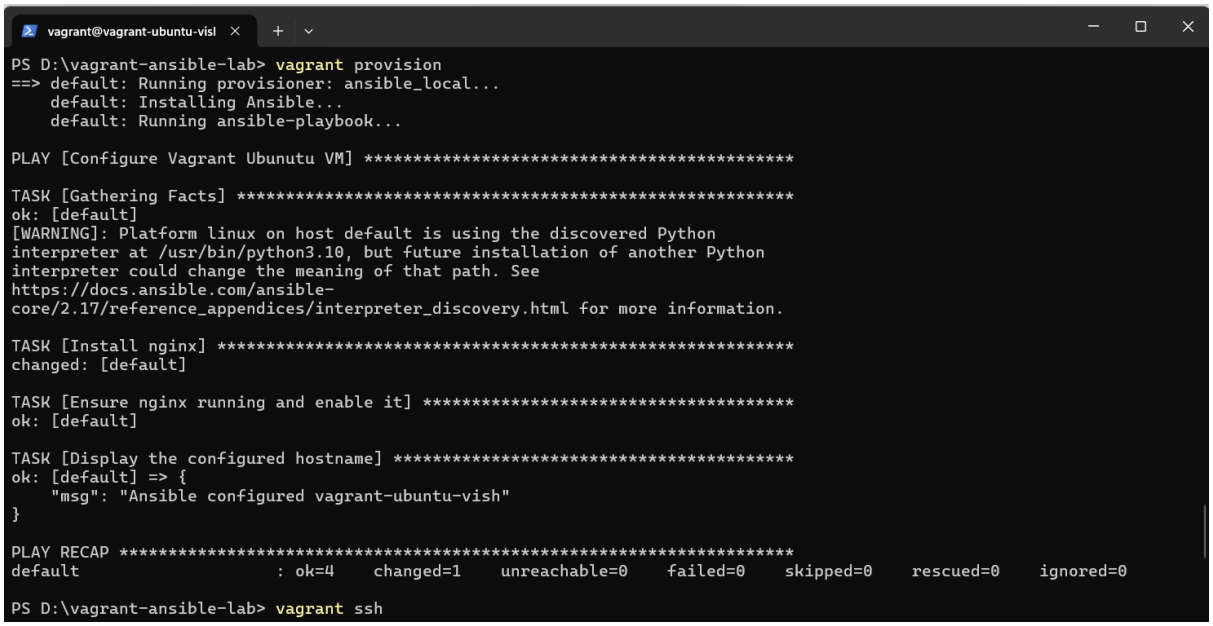

**There we go, successfully provisioned my first Ubuntu 22.04 VM using Vagrant + VMware Workstation, established secure SSH access, and verified the guest OS**



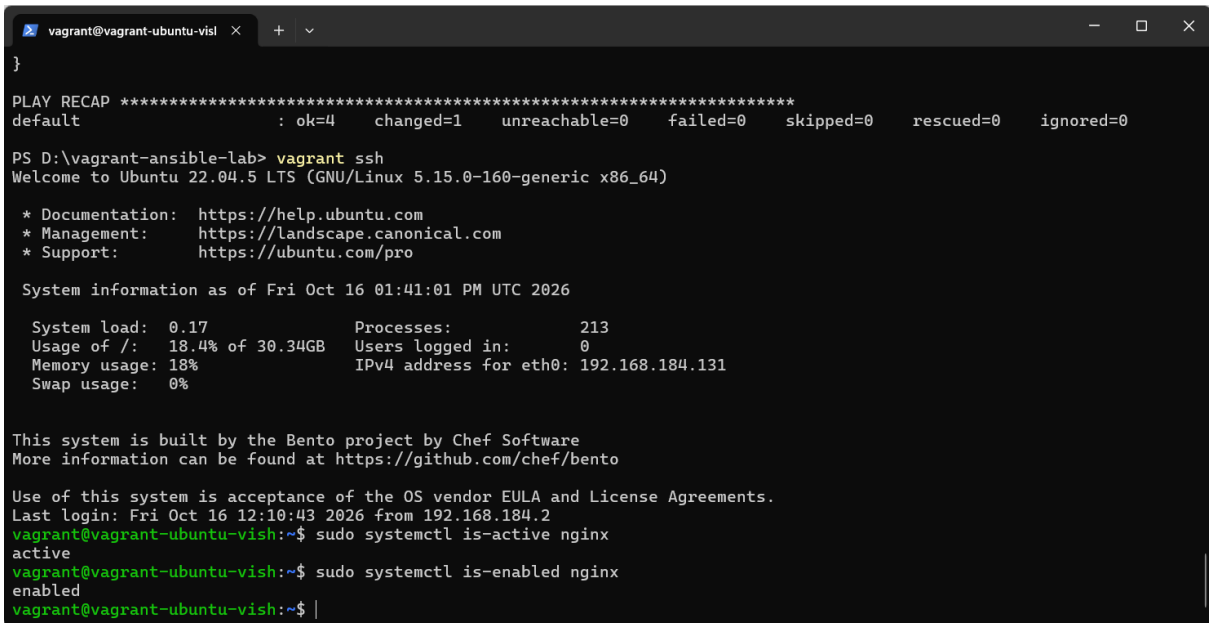

**Added Ansible Local Provisioning Step to the Vagrantfile**




**The ansible-playbook.yml file**




# Successful Ansible Local Provisioning: Nginx Installation and Service Configuration




# Finally, verified on the Ubuntu VM!!